# Exit Ticket · reflect and retrieve

## Exit Ticket

A variable that changes value inside a Karnaugh-map group is removed from that group's simplified term.

## Use key vocabulary from memory

1. Explain Karnaugh map in your own words.

2. Explain Group in your own words.

3. Explain Adjacent in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Karnaugh maps 1: start with four squares

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Write −13 as an 8-bit two’s-complement integer .

### Q2

For the circuit shown, simplify Q = ¬(A ∨ B) using De Morgan’s Law.

### Q3

Complete one full left-to-right pass of ascending bubble sort. What is the resulting array?

### Q4

Normalise this positive binary mantissa without changing its value. Give the new exponent.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Map the truth-table outputs — For each combination of A and B, find the same output Q in the map. The grid moves outputs; it does not change them.

## Core idea 2

Group adjacent 1s — Circle the two 1s in the bottom row. A group can contain 1, 2, 4, 8… adjacent 1s. Use the largest useful group.

## Core idea 3

Compare the input values — For the two circled cells: — CODE:
A = 1, B = 0
A = 1, B = 1

**Key visual · complete concept**

BC columns

| A rows | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 1 | 0 |

## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 2

Free-popcorn voucher
E = voucher shown, F = soft drink, G = water. Free popcorn is offered when E AND (F OR G). Circle two useful pairs and write R as an OR of two AND terms. Adapted from OCR Boolean Algebra Q2(b).

**Question 2 · original source diagram**

FG columns

| E rows | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 | 1 |

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

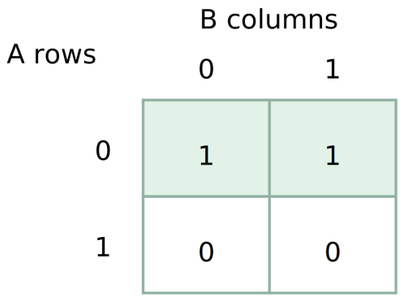
Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

One horizontal pair

Circle both adjacent 1s. For A and B, state which stays constant or changes. Write the simplified Q.

**Question 1 · original source diagram**




Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Core idea 4

Why Gray code? — From 01 to 10 , two bits change — these cannot be neighbouring K-map columns. — Each neighbouring pair differs by one bit only , including the end pair 10 ↔ 00 .

## Core idea 5

Read the three-input layout — Two bits label each BC column. Moving sideways changes one input only. This allows useful groups in three-variable maps.

## Vocabulary · use the correct technical terms

Karnaugh map — A grid used to simplify the outputs from a truth table.

Group — Adjacent 1s circled as one rectangular region.

Adjacent — Cells next to each other horizontally or vertically.

Simplify — Express the same output with fewer Boolean variables.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

How does this four-square map simplify?

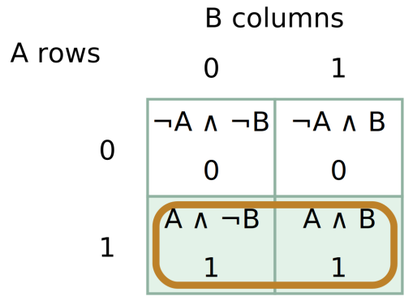
## Follow the worked method

1. Read the input combinations — Only the two bottom cells contain 1. They represent A=1, B=0 and A=1, B=1 .

2. Compare each input

3. Write the simplified expression — Keep A because it remains 1. Remove B because it changes between 0 and 1. — Check: Q is 1 for both values of B whenever A = 1.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

1. Identify the group
Circle the two adjacent 1s.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.